UNIVERSITY OF TWENTE.

Performance Evaluation for Collision Prevention

based on a Domain Specific Language

Freek van den Berg, Anne Remke,
Arjan Mooij and Boudewijn Haverkort

COMMIT/ALLEGIO
Embedded Systems Innovation BY TNO
PHILIPS sense and simplicity

Overview

- Collision prevention
- Domain Specific Language (DSL)
- Performance model
- Model validation

Collision prevention case study

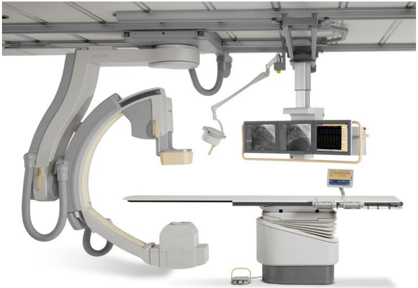
iXR

- Embedded systems
  - increased complexity
- Early performance prediction
  - before the system exists

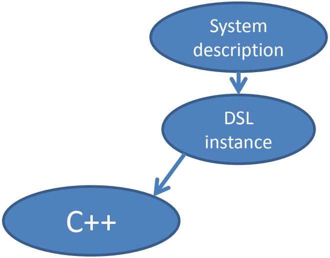
Collision prevention case study

iXR

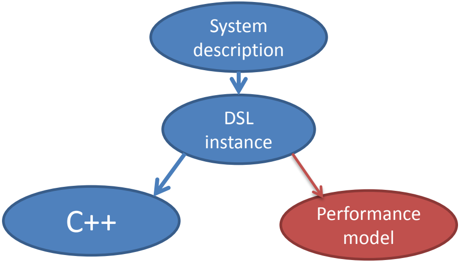
- Embedded systems
  - increased complexity
- Early performance prediction
  - before the system exists
- Collision prevention in interventional X-Ray machines



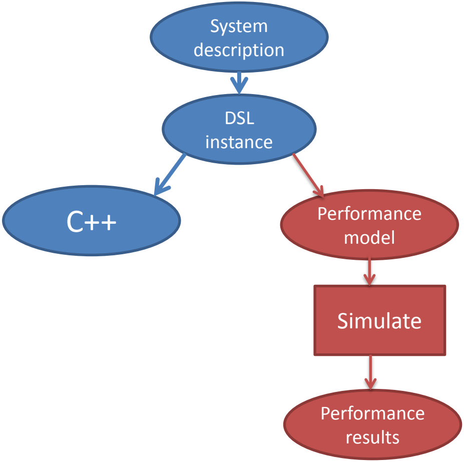
Domain specific language (DSL)



Domain specific language (DSL)



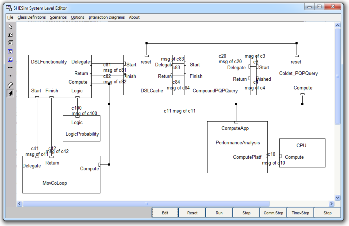
Domain specific language (DSL)



Parallel Object-Oriented Specification Lang.

... as the language of our performance model

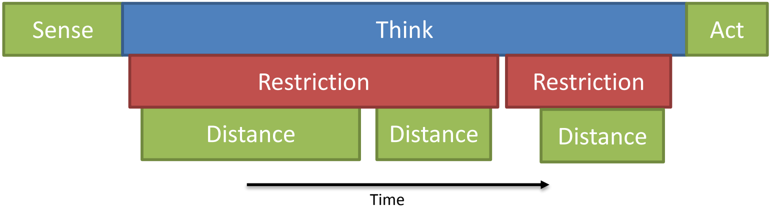
- Based on Stochastic Timed Automata (STA)
  - **Simulation**
  - Markov Chain based
  - **Expressive**
  - Arbitrary distributions



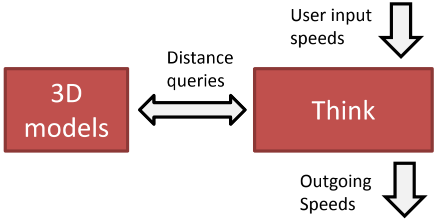
The Movement Control Loop

... implements Collision Prevention

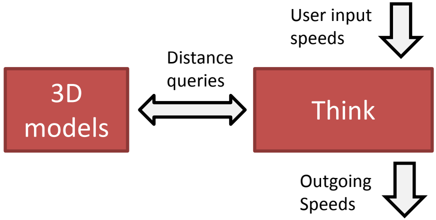
Sense	**Think**	**Act**
-receive positions of all objects	restrictions - derive speed limits - compute distances	-enforce speed limits on all objects



Domain Specific Language (DSL) Instance

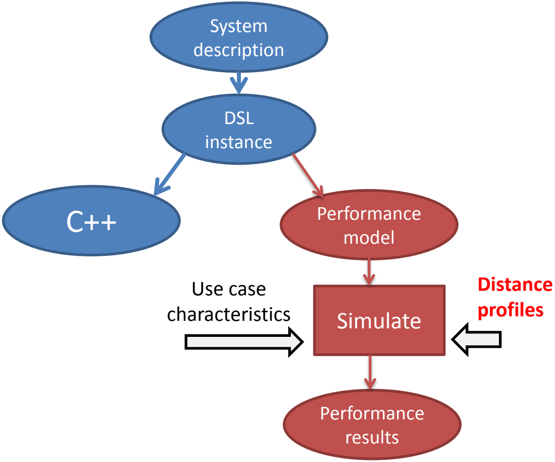


Domain Specific Language (DSL) Instance




```
restriction ApproachingTableTopArc
activation
   Distance[Future](TableTop, Arc) <133 &&
   Distance[Future](TableTop, Arc) <Distance[Now](TableTop, Arc)
effect
   limit Arc speed at (Distance[Future](TableTop, Arc) / 100))
```

Domain specific language (DSL)



Distance profiles

... define the execution time of distance queries

The performance of distance computations depends on

- the object complexities (number of triangles)
    - Profile per **object pair**

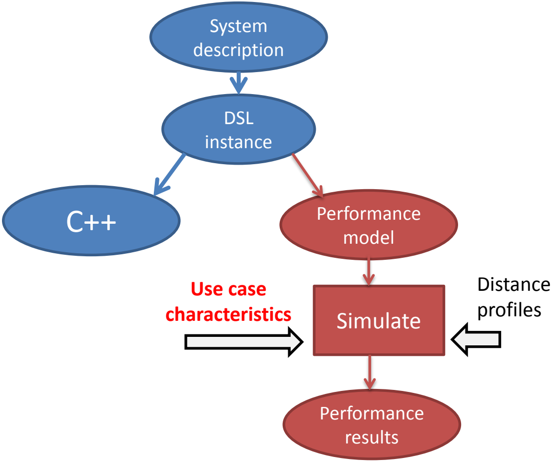
Distance profiles

... define the execution time of distance queries

The performance of distance computations depends on

- the object complexities (number of triangles)
  - Profile per **object pair**
- the relative geometric positions of the objects
  - Use **four profiling strategies**
    - Typical system usage (2D and 3D)
    - Broad scan (9D grid and random)

Domain specific language (DSL)



Use case characteristics

... define the **probabilities**
that distance queries get executed

```
restriction ApproachingTableTopArc
activation
   Distance[Future](TableTop, Arc) <133 &&
   Distance[Future](TableTop, Arc) <Distance[Now](TableTop, Arc)
effect
   limit Arc speed at (Distance[Future](TableTop, Arc) / 100))
```

Conditional distances result from
lazy and conditional evaluation

Use case characteristics

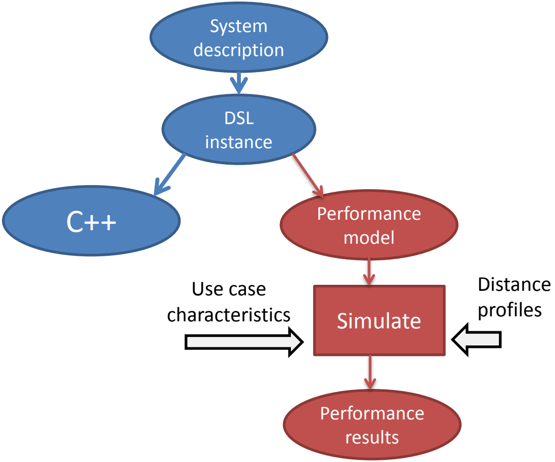
... define the **probabilities**
that distance queries get executed

```
restriction ApproachingTableTopArc
activation
  Distance[Future](TableTop, Arc) <133 &&
  Distance[Future](TableTop, Arc) <Distance[Now](TableTop, Arc)
effect
  limit Arc speed at (Distance[Future](TableTop, Arc) / 100))
```

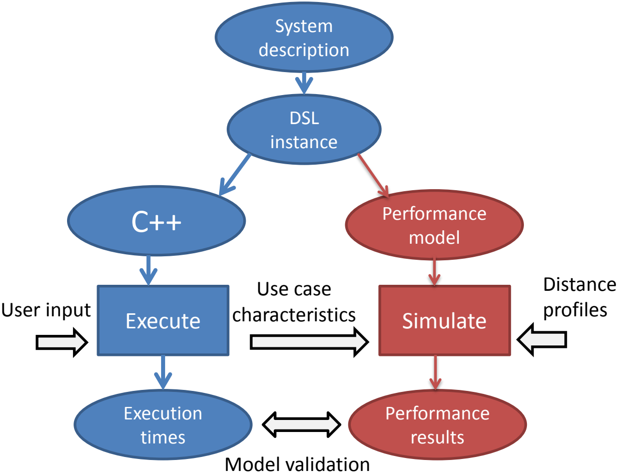
Conditional distances result from
lazy and conditional evaluation

	red	blue
UC Table moves	0.07	0.04
UC Arc moves	0.29	0.99
UC Stationary	0.01	0.79

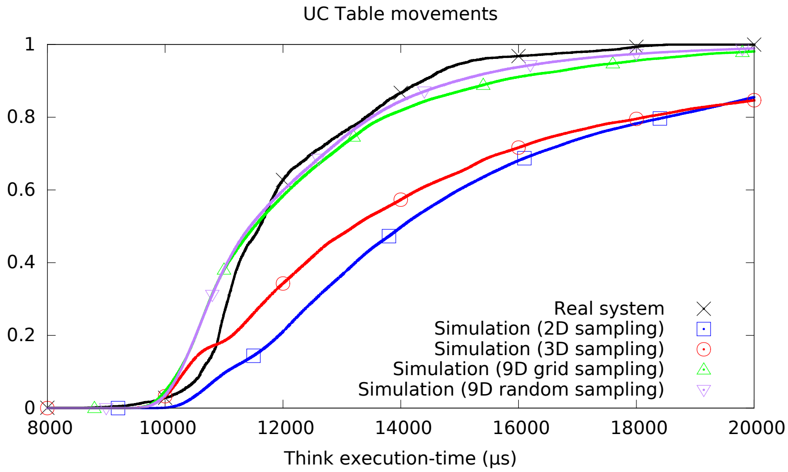
Model validation



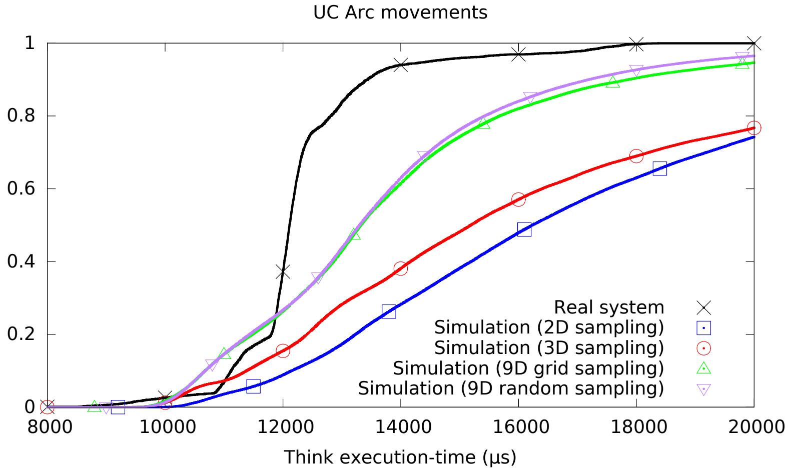
Model validation



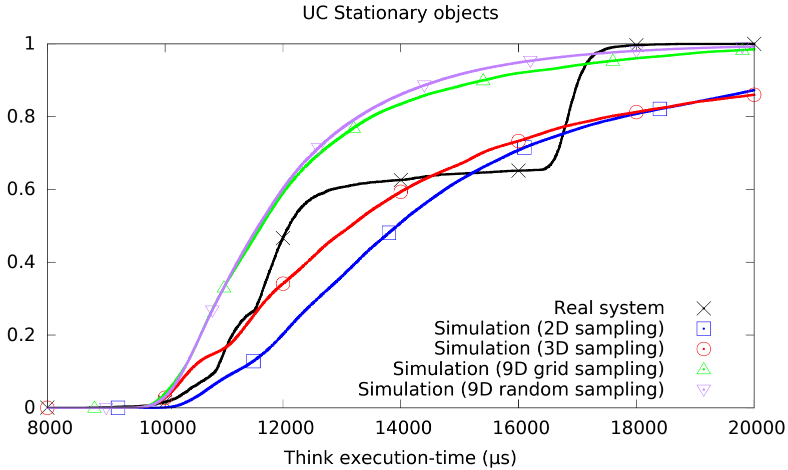
Model validation: Execution-time CDFs



Model validation: Execution-time CDFs



Model validation: Execution-time CDFs



Take home messages

- DSL-based performance evaluation
  - DSL automatically transforms into the performance model
- Our approach
  - provides reasonable performance estimates
  - can be used **early** in design
  - is **efficient** to implement when a DSL is already present

Thank you for your attention